

Find the wrong term in the following series.

Q1: 12, 11, 16, 28, 50, 85

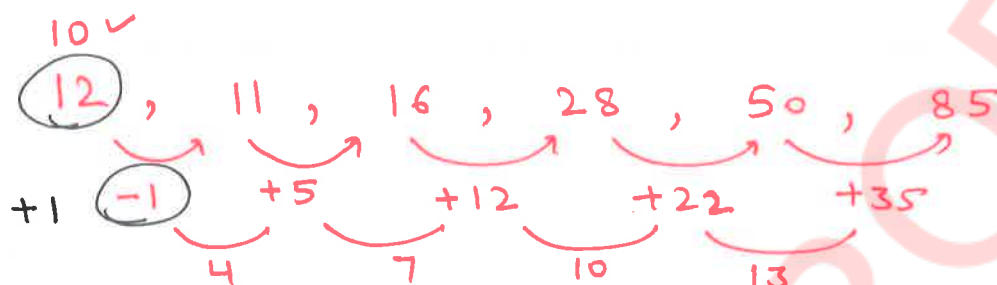
(A) 12

(B) 11

(C) 16

(D) 28

(E) 50



Q2: 2, 5, 24, 120, 720, 5040

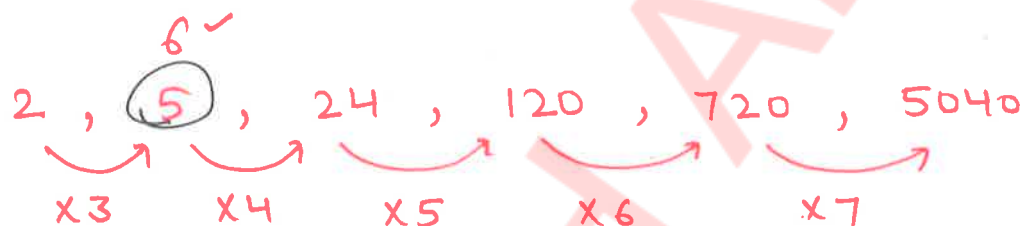
(A) 5

(B) 24

(C) 120

(D) 720

(E) 5040



Q3: 1, 12, 123, 1234, 12345, 123457

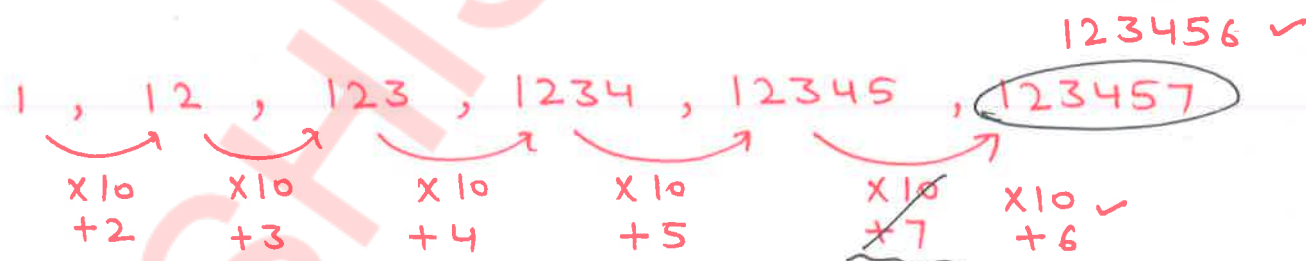
(A) 12

(B) 123

(C) 1234

(D) 12345

(E) 123457



Q4: 9, 18, 72, 432, 3458, 34560

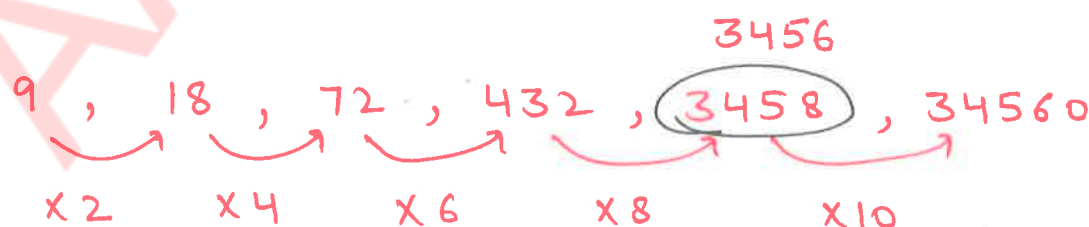
(A) 432

(B) 3458

(C) 34560

(D) 72

(E) 18



Q5: 81, 84, 78, 87, 75, 93

(A) 87

(B) 93

(C) 75

(D) 78

(E) 81

81, 84, 78, 87, 75, 93

$+3$ -6 $+9$ -12 $+18$
 $+15$

75 + 15 = 90 ✓

Q6: 243, 162, 216, 432, 1152, 1149

(A) 162

(B) 216

(C) 432

(D) 1152

(E) 1149

243, 162, 216, 432, 1152, 1149

$\times \frac{2}{3}$ $\times \frac{4}{3}$ $\times \frac{6}{3}$ $\times \frac{8}{3}$ $\times \frac{10}{3}$

$1152 \times \frac{10}{3}$
 $= 384 \times 10$
 $= 3840 \checkmark$

Q7: 20, 39, 74, 139, 261, 499

(A) 39

(B) 261

(C) 139

(D) 20

(E) 499

20, 39, 74, 139, 261, 499

$\times 2 - 1$ $\times 2 - 4$ $\times 2 - 9$ $\times 2 - 17$ $\times 2 - 23$
 $\times 2 - 16$ $\times 2 - 25$

262 ✓

Q8: 4, 10, 18, 23, 46, 51

(A) 10

(B) 18

(C) 23

(D) 46

(E) 51

4, 10, 18, 23, 46, 51

$+5$ $\times 2$ $+5$ $\times 2$ $+5$

9 ✓

$n!$ → The product of all positive integers from n upto 1.

$n!$ → $n \times (n-1) \times (n-2) \times (n-3) \times \dots \times 1$.

$$1! = 1$$

$$2! = 2 \times 1 = 2$$

$$3! = 3 \times 2 \times 1 = 6$$

$$4! = 4 \times 3 \times 2 \times 1 = 24$$

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

$$6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$$

Q9: 3, 4, 6, 12, 35, 156

(A) 4

(B) 6

(C) 12

(D) 35

(E) 156

36 ✓

3, 4, 6, 12, 35, 156

$\xrightarrow{+1}$ $\xrightarrow{+2}$ $\xrightarrow{+6}$ $\xrightarrow{+24}$ $\xrightarrow{+121}$ $\xrightarrow{+120}$

$\xrightarrow{\times 2}$ $\xrightarrow{\times 3}$ $\xrightarrow{\times 4}$ $\xrightarrow{\times 5}$

Q10: 2, 12, 30, 68, 130, 222

(A) 12

(B) 30

(C) 68

(D) 130

(E) 222

10

2, 12, 30, 68, 130, 222

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$

$(1+1)$ $(8+2)$ $(27+3)$ $(64+4)$ $(125+5)$ $(216+6)$

(1^3+1) (2^3+2) (3^3+3) (4^3+4) (5^3+5) (6^3+6)

Q11: 2, 4, 7, 13, 27, 53

(A) 4

(B) 7

(C) 13

(D) 27

(E) 53

3 ✓

2, 4, 7, 13, 27, 53

$\xrightarrow{\times 2}$ $\xrightarrow{\times 2}$ $\xrightarrow{\times 2}$ $\xrightarrow{\times 2}$ $\xrightarrow{\times 2}$

-0 -1 -1 $+1$ -1

$\underline{\times 2 - 1}$ $\underline{\times 2 + 1}$

Q12: 4, 8, 18, 32, 50, 72

(A) 4

(B) 8

(C) 18

(D) 32

(E) 72

$\overset{2\checkmark}{\textcircled{4}}, 8, 18, 32, 50, 72$
 $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $(1 \times 2) \quad (4 \times 2) \quad (9 \times 2) \quad (16 \times 2) \quad (25 \times 2) \quad (36 \times 2)$
 $(1^2 \times 2) \quad (2^2 \times 2) \quad (3^2 \times 2) \quad (4^2 \times 2) \quad (5^2 \times 2) \quad (6^2 \times 2)$

Q13: 11, 8, 24, 21, 64, 60

(A) 11

(B) 8

(C) 21

~~(D) 64~~

(E) 60

$11, 8, 24, 21, \textcircled{64}, 60$
 $\underbrace{\quad} \quad \underbrace{\quad} \quad \underbrace{\quad} \quad \underbrace{\quad}$
 $-3 \quad \times 3 \quad -3 \quad \times 3 \quad -3$

Q14: 52, 106, 222, 459, 939, 1906

~~(A) 52~~

(B) 106

(C) 222

(D) 459

(E) 939

$$2x + 6 = 106$$

$$2x = 100$$

$$x = 50$$

$\overset{50}{\textcircled{52}}, 106, 222, 459, 939, 1906$
 $\underbrace{\quad} \quad \underbrace{\quad} \quad \underbrace{\quad} \quad \underbrace{\quad}$
 $\times 2 \quad \times 2 \quad \times 2 \quad \times 2 \quad \times 2$
 $+6 \quad +10 \quad +15 \quad +21 \quad +28$
 $\underbrace{\quad} \quad \underbrace{\quad} \quad \underbrace{\quad} \quad \underbrace{\quad}$
 $+4 \quad +5 \quad +6 \quad +7$

Q15: 60, 85, 170, 219, 438, 520

(A) 85

(B) 170

(C) 219

(D) 438

~~(E) 520~~

$60, 85, 170, 219, 438, \textcircled{520}^{\overset{519}{}}$
 $\underbrace{\quad} \quad \underbrace{\quad} \quad \underbrace{\quad} \quad \underbrace{\quad}$
 $+25 \quad \times 2 \quad +49 \quad \times 2 \quad +81$
 $(5^2) \quad (7^2) \quad (9^2)$

Q16: 5, 9, 36, 52, 176, 213

(A) 9

(B) 36

(C) 52

~~(D) 176~~

(E) 213

$$\begin{array}{cccccc}
 5 & , & 9 & , & 36 & , & 52 & , & \textcircled{176} & , & 213 \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 +4 & & +27 & & +16 & & +124 & & +37 \\
 (2^2) & & (3^3) & & (4^2) & & (5^3) & & (6^2)
 \end{array}$$

Q17: 5, 16, 45, 135, 405, 1215, 3645

~~(A) 16~~

(B) 45

(C) 135

(D) 405

(E) 1215

$$\begin{array}{cccccc}
 5 & , & \textcircled{16} & , & 45 & , & 135 & , & 405 & , & 1215 & , & 3645 \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 \times 3 & & \times 3 & & \times 3 & & \times 3 & & \times 3 & & \times 3 & & \times 3
 \end{array}$$

Q18: 15015, 10010, 6006, 4290, 2730, 2305

(A) 10010

(B) 6006

(C) 4290

(D) 2730

(E) 2305

$$\begin{array}{cccccc}
 15015 & , & 10010 & , & 6006 & , & 4290 & , & 2730 & , & \textcircled{2305} & \xrightarrow{2310} \checkmark \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 \times \frac{2}{3} & & \times \frac{3}{5} & & \times \frac{5}{7} & & \times \frac{7}{11} & & \times \frac{11}{13}
 \end{array}$$

$$\left\{ 2730 \times \frac{11}{13} = 2310 \right\}$$

Q19: 7, 10, 39, 140, 725, 4315

(A) 140

(B) 725

~~(C) 4315~~

(D) 39

(E) 10

$$\begin{array}{cccccc}
 7 & , & 10 & , & 39 & , & 140 & , & 725 & , & \textcircled{4315} \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 \times 2 & & \times 3 & & \times 4 & & \times 5 & & \times 6 \\
 -4 & & +9 & & -16 & & +25 & & -36
 \end{array}$$

$$\begin{aligned}
 & (725 \times 6) - 36 \\
 & 4350 - 36 \\
 & = \underline{4314}
 \end{aligned}$$

Q20: 7, 11, 29, 83, 245, 731

(A) 7

(B) 11

(C) 29

(D) 83

(E) 245

$(x)3 - 4 = 11$
 $3x = 15$
 $x = 5$

5 ✓
 7, 11, 29, 83, 245, 731

$\begin{array}{r} \times 3 \\ -10 \\ \hline \times \end{array}$
 $\begin{array}{r} \times 3 \\ -4 \\ \hline \end{array}$
 $\begin{array}{r} \times 3 \\ -4 \\ \hline \end{array}$
 $\begin{array}{r} \times 3 \\ -4 \\ \hline \end{array}$
 $\begin{array}{r} \times 3 \\ -4 \\ \hline \end{array}$

Q21: 100, 52, 28, 16, 10, 5

(A) 52

(B) 28

(C) 16

(D) 10

(E) 5

$2(x) - 4 = 10$
 $2x = 14$
 $x = 7$

100, 52, 28, 16, 10, 5

$\begin{array}{r} \times 2 \\ -4 \\ \hline \end{array}$
 $\begin{array}{r} \times 2 \\ -4 \\ \hline \end{array}$
 $\begin{array}{r} \times 2 \\ -4 \\ \hline \end{array}$
 $\begin{array}{r} \times 2 \\ -4 \\ \hline \end{array}$
 $\begin{array}{r} \times 2 \\ -0 \\ \hline \times \end{array}$

Q22: 6, 8, 17, 44, 125, 368

(A) 6

(B) 8

(C) 17

(D) 44

(E) 125

$3(x) - 7 = 8$
 $3x = 15$
 $x = 5$

5, 6, 8, 17, 44, 125, 368

$\begin{array}{r} \times 3 \\ -10 \\ \hline \times \end{array}$
 $\begin{array}{r} \times 3 \\ -7 \\ \hline \end{array}$
 $\begin{array}{r} \times 3 \\ -7 \\ \hline \end{array}$
 $\begin{array}{r} \times 3 \\ -7 \\ \hline \end{array}$
 $\begin{array}{r} \times 3 \\ -7 \\ \hline \end{array}$

Q23: 75, 107, 171, 299, 555, 1069

(A) 107

(B) 171

(C) 299

(D) 555

(E) 1069

75, 107, 171, 299, 555, 1069 ✓

32, 64, 128, 256, 512

$\begin{array}{r} \times 2 \\ -514 \\ \hline \end{array}$

Q24: 3, 6, 8, 13, 21, 34

(A) 6

(B) 8

(C) 13

(D) 21

(E) 34

$3, \overset{5}{\textcircled{6}}, 8, 13, 21, 34$
 $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $(3+5) \quad (5+8) \quad (8+13) \quad (13+21)$

{ Sum of previous two terms }
Fibonacci Pattern

Q25: 5, 14, 51, 220, 1125, 6788

(A) 220

(B) 1125

(C) 6788

(D) 51

(E) 14

$5, 14, 51, 220, 1125, \overset{6786}{\textcircled{6788}}$
 $\begin{array}{cccccc} \text{---} & \text{---} & \text{---} & \text{---} & \text{---} & \text{---} \\ \times 2 & \times 3 & \times 4 & \times 5 & \times 6 & \\ +4 & +9 & +16 & +25 & +36 & \end{array}$
 $(1125 \times 6) + 36$
 $6750 + 36$
6786 ✓

Q26: 30, 27, 54, 51, 104, 99

(A) 27

(B) 54

(C) 51

(D) 104

(E) 99

$30, 27, 54, 51, \overset{102}{\textcircled{104}}, 99$
 $\begin{array}{cccccc} \text{---} & \text{---} & \text{---} & \text{---} & \text{---} & \text{---} \\ -3 & \times 2 & -3 & \times 2 & -3 & \end{array}$

$\begin{cases} 51 \times 2 = 102 \\ 102 - 3 = 99 \end{cases}$

Q27: 22, 23, 25, 26, 27, 28

(A) 23

(B) 25

(C) 26

(D) 27

(E) 28

22, ²⁴
 22, 23, 25, 26, 27, 28

- 23 is a prime Number
- All other numbers are consecutive composite numbers.
- Next composite number after 22 = 24

Q28: 13, 14, 18, 26, 38, 62

(A) 13

(B) 14

(C) 18

(D) 26

(E) 62

12
13, 14, 18, 26, 38, 62

$\xrightarrow{+2}$ $\xrightarrow{+4}$ $\xrightarrow{+8}$ $\xrightarrow{+12}$ $\xrightarrow{+24}$
 (1x2) (1x4) (1x8) (2x6) (3x8)

$$\Rightarrow \begin{aligned} 12 + (1 \times 2) &= 14 \\ 14 + (1 \times 4) &= 18 \\ 18 + (1 \times 8) &= 26 \\ 26 + (2 \times 6) &= 38 \\ 38 + (3 \times 8) &= 62 \end{aligned}$$